

Home Assignment: Scene Description via VLM

The Dataset

Use the [nuScenes v1.0-mini dataset](#).

```
pip install nusenes-devkit
```

See [this notebook](#) for how to work with it.

The Task

Given a set of driving scenes, run each through a vision-language model (VLM) and output a scene description.

Build a small pipeline that:

1. **Loads a scene** — pick a representative keyframe image (e.g. the front camera) from each scene.
 1. **Runs a VLM** — pass the scene through a VLM and produce a short natural-language description of the scene.
 1. **Outputs the results** — write the descriptions out in a simple structured form (e.g. JSON or a table), one entry per scene.
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What We're Looking For

- How did you test the code?
 - Even though this could be a simple script, structure it as if this was a production project.
 - How would you deploy this project
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Expectations

- Include a README on how to set up and run the code.
 - The code should be containerized.
 - Python is recommended, but use any language you like.
 - State your assumptions explicitly.
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Notes

- **Use any tools you want** — ie Claude, Codex etc.
- No need for large models or GPU inference. A small/basic VLM is fine even if the descriptions aren't perfect.